

Exercise 1: Rates

1. Which of these are rates?

- | | |
|--------------------|-------------|
| (a) km/h | (b) \$/kg |
| (c) c/L | (d) L/min |
| (e) 4 h 25 min | (f) kbps |
| (g) \$120 per week | (h) \$62.50 |
| (i) 35 wpm | (j) 8% p.a. |

2. Give an example of a **rate** that could be used in relation to;

- | | |
|----------------------------|---------------|
| (a) a hospital | (b) a car |
| (c) computers | (d) a house |
| (e) a fruit and vegie shop | (f) a job |
| (g) a timber yard | (h) an office |

3. Where would these **rates** be used?

- | | |
|-----------------|----------------------------|
| (a) \$/h | (b) \$/year |
| (c) L/100km | (d) \$/m |
| (e) % per annum | (f) words per minute (wpm) |
| (g) m/s | (h) bps |

4. Complete these tables at the given **rate**:

(a) A car travels at 100km/h

Time in hours	1	2	3	5		14
Distance in km					650	

(b) An average reader reads at 250 words per minute

Time in min	1	2	5	10		30
Words read					4000	

(c) House rental is \$145 per week

Weeks rented	1	2	4	10		52
Cost in dollars					3190	

(d) Petrol cost 76c/L

Litres bought	1	5	10	50		80
Cost in dollars					\$54.72	

(e) Jayde gets paid \$8.60 per hour

Hours worked	$\frac{1}{2}$	1	2	$3\frac{1}{2}$		8
Pay in dollars					\$43	

5. Calculate the unit **rate** required in each of these;

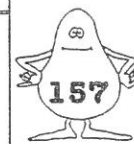
- \$/kg if \$40 buys 8kg worth of peaches
- wpm if Janis types 420 words in 7 minutes
- \$/m² if it costs \$250 to pave 16 square metres
- m/s if a roller coaster travels 220m in 7 seconds
- \$/m if 15m of timber costs \$61.50
- m/\$ if 15m of timber costs \$61.50
- L/min if a pool pump moves 4.5kL of water in 14 minutes
- L/100km if a car uses 28L to go 350km
- c/L if 55 litres costs \$39.60
- \$/week if 12 weeks rental costs \$1620



6. Joshua is getting paid \$350 to work a 40 hour week during the holidays.
- (a) What is his pay **rate** in \$/h?
 - (b) How much would he be paid for 6 hours?
 - (c) How much would he be paid for 15 hours?
 - (d) How much would he be paid for $3\frac{1}{2}$ hours?
 - (e) How many hours would he have to work to earn \$175?
 - (f) How many hours would he have to work to earn \$280?
 - (g) How many hours would he have to work to earn \$41.56?
 - (h) Is this a better pay **rate** than Paul's 24 hour week that earns him \$200?
7. If Pippa buys \$4 worth of chocolate coated sultanas at the pictures and the bag weighs 1.25kg;
- (a) What is their price in \$/kg?
 - (b) How many grams does she get per dollar?
 - (c) How much would be in a \$5 bag?
 - (d) How much would be in a \$10 bag?
 - (e) How much would it cost for a 2kg bag?
 - (f) How much would it cost for a $1\frac{1}{2}$ kg bag?
8. Lauren's parents have a Commodore that has a fuel consumption **rate** of 11.5L/100km on city cycle and 7.5L/100km on highway cycle.
- (a) If they purchase petrol at 82.5c/L how much will it cost them for fuel to travel 450 km in the country (highway cycle)?
 - (b) How far could they travel in the country with a 55L fuel tank?
 - (c) If they purchase petrol at 75.5c/L how much will it cost them for fuel to travel 450km in the city (city cycle)?
 - (d) How far could they travel in the city with a 55L fuel tank?



WARNING TO ESKIMOS



ANSWER THE QUESTIONS
BELOW (REGARDING SPEED)
TO FIND THE PUZZLE
SOLUTION CODE.



How far will a car go at
60 km/h for 7 hours?



r



4 m/s

How far will
the skateboard
go in 21 sec?

a

How long will it take a
bird to fly 200 km, flying
at 40 km/h?



n

How long 70 m/min
will it take
this swimmer
to go 630 m?



o



600 km/h

How far will
the jet go in $2\frac{1}{2}$ hr?

v

How far will a boat moving
at 40 km/h go in 45 min?



y

The dog runs at
20 m/s for 45
sec. How far will
it have run?



e



A snail's
pace is 3 m/h

How far will it travel in
a day?

g

A cricket ball
moving at
25 m/sec takes 5 sec to hit
a fence. How far did it
travel?



i



A pram was pushed at
2 km/h for 15 min.
How far did it go?

u

A bee moving at 16 m/s
covers 112 m. How long
will this take?



h

A bicycle moving
at 35 km/h travels
for 3 hours. How far
did it go?



p

A shark
swimming
at 5 m/s covers 120 m in
what time?



d



A street cleaner
sweeps gutters
at 8 m/min. How
long will it take to
cover 960 m?

s



Water flows
down a creek
at a rate of
21 m/min.
How far will
a toy yacht go
in 10 min?

t

A rocket travels at
150 m/s. How far
will it move in one
minute?



m



How far will
a cloud move
in 30 minutes if it is
travelling at 50 m/min?

w

24 sec	9 min	5 h	210 m	72 m	125m	1500 km	900 m	84 m			
7 sec	9 min	500 m	2 h	900 m	1500 m	84 m	420 km	9 km	125 m	5 h	72 m
105 km	84 m	420 km	210 m	30 km	210 m	9 min	24 sec	84 m	30 km		